NIHVA Technologies Pvt Ltd



We Make Possible

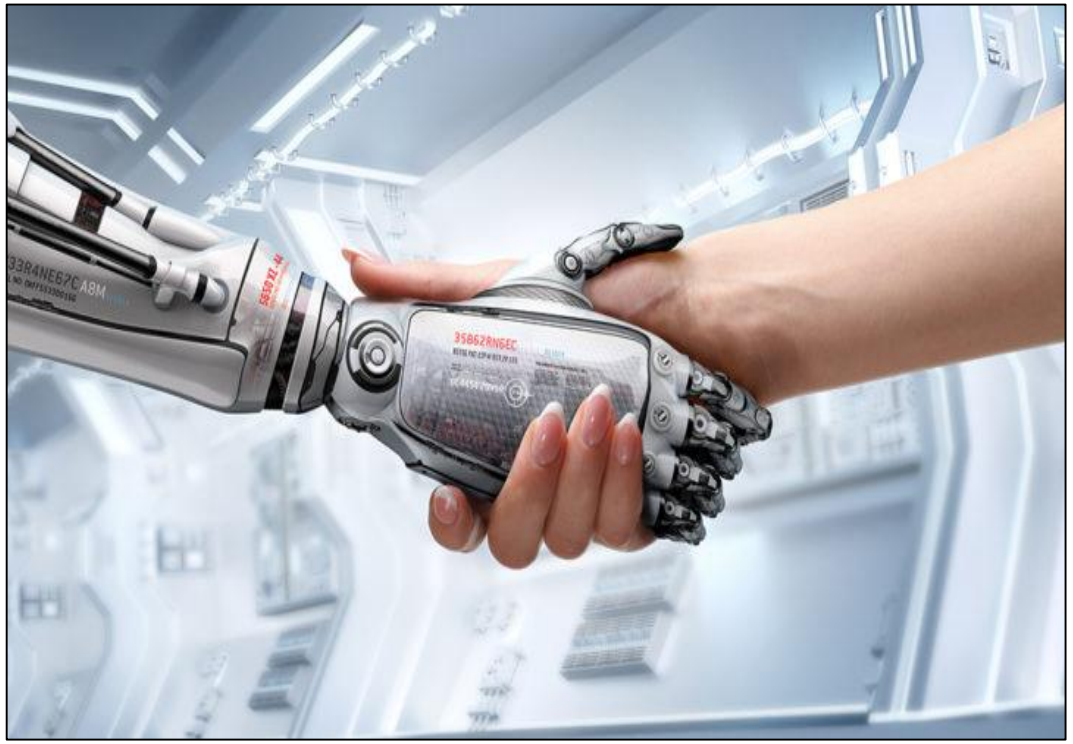
NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

NIHVA Major Domains:

Door Automation



Automation Solutions



NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Client's Requirement

Product: Motorized Retractable Roof

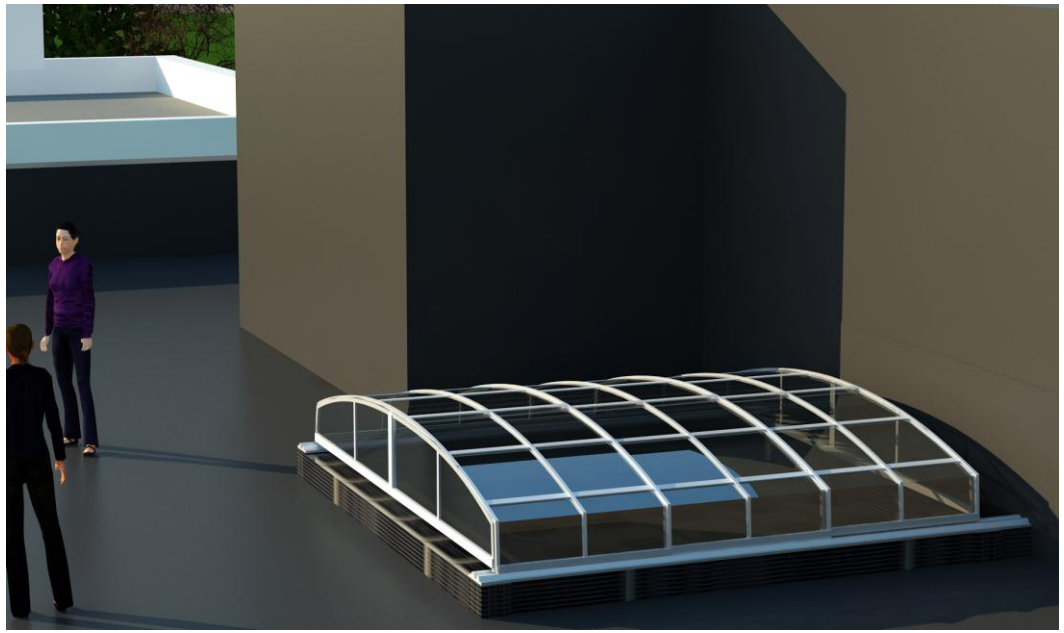
NIHVA
NIHVA Technologies Pvt. Ltd.

Sr. No	Size	Quantity in NO's
1	Size (W) 2400 mm x (L) 2200 mm x (H) 400 mm x 2 panels	9
2	Size (W) 2600 mm x (L) 2400 mm x (H) 400 mm x 2 panels	12
3	Size (W) 3000 mm x (L) 6000 mm x (H) 600 mm x 3 panels	1
4	Size (W) 3200 mm x (L) 11500 mm x (H) 700 mm x 4 panels Size (W) 5200 mm x (L) 19000 mm x (H) 850 mm x 4 panels	1
5	Size (W) 2500 mm x (L) 10900 mm x (H) 700 mm x 4 panels Size (W) 4000 mm x (L) 3200 mm x (H) 650 mm x 2 panels Size (W) 5050 mm x (L) 11800 mm x (H) 900 mm x 4 panels	1
6	Size (W) 2500 mm x (L) 11300 mm x (H) 700 mm x 4 panels	1
7	Size (W) 9200 mm x (L) 14000 mm x (H) 1000 mm x 4 panels	1
8	Size (W) 2600 mm x (L) 8300 mm x (H) 700 mm x 4 panels Size (W) 3800 mm x (L) 14000 mm x (H) 750 mm x 4 panels	1

Brief Introduction: Motorized Retractable Roof

NIHVA Technologies Pvt. Ltd.

Retractable Roof is kind of innovation for Bungalows, Hotels, Hospital, Villas, Apartments, commercial complex, Offices, where we can add more glory by allowing partial sunlight to enter in premises at day and start sky view at nights. Roofs are operated in many ways you like Folding Roof, sliding roof, telescopic roof, single sided, double sided, vertical lift, inclined lift and etc.



Technical Specifications of Motorized Retractable Roof

NIHVA
NIHVA Technologies Pvt. Ltd.

Sr. No	Description	Specification
1	PC Sheet	3.0 mm Thick
2	Sliding Roof Material	Powder Coated Aluminum with Polycarbonate
3	Profile	White RAL Powder Coated AL
4	Guide	Anodized AL Guide
5	Rain Guard and Gutter	Rain Guard and Gutter
6	Drive	Rack and Pinion Mechanism
7	Operating type	Single (To & Fro motion) Sliding Type

Roof Material Details

NIHVA
NIHVA Technologies Pvt. Ltd.

1. Aluminium profiles and tracks.

- The retractable roof is constructed with the combination of the 6060 grades Non corrosive Aluminium hollow extruded profiles 60x 63x2, 63.3x 106.75x2.5,50x25x2.
- Aluminium extruded tracks 14.8x94.5x2.35, 14.8x134x2.35.
- Alloy- EN-AW-6060-T6 as per BSEN755.

2. Pre-treatment and finishing.

2.1 Surface preparation

- A multistage cleaning and pre-treatment system are adopted to remove the organic and inorganic surface soils, remove residual oxides and apply a Chrome chemical conversion coating to which the organic coatings will firmly adhere. This is done as per ASTM D 1730- 09.

2.2 Powder coating

- The prepared aluminium is powder coated with Type Super Durability powder coat as per standard specifications: Qualicoat Class 2 AAMA 2604.

2.3 Sealed Anodic oxidation coatings:

- Aluminium is coated with anodic oxide coating thickness to Grade AC25 as per IS 1868 & sealing ISO-3210 standards.

3. Poly carbonate – Double sided UV protected sheets

- Polycarbonate sheets with UV protection on both sides. Sheets resist temperatures of -100 to +120 °C.

NIHVA
NIHVA Technologies Pvt. Ltd.

Description	Test Conditions	Typical values (1)	Unit	Standard
PHYSICAL				
Density		1200	kg/m3	ISO 1183-1
Water absorption saturation	water at 23 °C	0.35	%	ISO 62-1
Water absorption equilibrium	23 °C, 50% relative humidity	0.15	%	ISO 62-4
MECHANICAL				
Tensile stress at yield		› 60	MPa	ISO 527-2/1B/50
Elongation at yield		6	%	ISO 527-2/1B/50
Tensile strength		› 60	MPa	ISO 527-2/1B/50
Elongation at break				ISO 527-2/1B/50
Elastic Modulus		2400	MPa	ISO 527-2/1B/1
Limiting flexural stress		approx. 90	MPa	ISO 178

Description	Test Conditions	Typical values (1)	Unit	Standard
Impact strength	Charpy, unnotched	no break	kJ/m2	ISO 179/1fU
	Charpy, notched	approx. 11	kJ/m2	ISO 179/1eA
	Izod, notched	approx. 10	kJ/m2	ISO 180/1A
	Izod, notched1	approx. 70	kJ/m2	ISO 180/4A
THERMAL				
Vicat softening temperature	Method B50	148	°C	ISO 306
Thermal conductivity		0.2	W/mK	DIN 52612
Coeff. of linear thermal expansion		0.065	mm/m°C	DIN 53752-A
Heat deflection temperature under load	Method A: 1.80 MPa	127	°C	ISO 75-2
	Method B: 0.45 MPa	139	°C	ISO 75-2

NIHVA

NIHVA Technologies Pvt. Ltd.

NIHVA
NIHVA Technologies Pvt. Ltd.

Description	Test Conditions	Typical values (1)	Unit	Standard
ELECTRICAL				
Dielectric strength		35	kV/mm	IEC 60243-1
Volume resistivity		1016	Ohm·cm	IEC 60093
Surface resistivity		1014	Ohm	IEC 6009
Dielectric constant	at 103 Hz	3.1		IEC 60250
	at 106 Hz	3		IEC 60250
Dissipation factor	at 103 Hz	0.0005		IEC 60250
	at 106 Hz	0.009		IEC 60250
Light transmission Typical value		50	%	DIN 5036

Weathering Resistance: Excellent weathering resistance, which guarantees their impact strength even after many years. The sheets are provided with a 10-year warranty on weathering resistance, mechanical properties and on optical properties.

Four types of Polycarbonate Sheets are offered.

Compact Clear
Compact Bronze Clear
Clear Embossed
Bronze Embossed

Stainless steel fasteners

High grade Stainless steel SS304 fasteners are used to work against the harsh environmental conditions**.**

Fabricated Parts

All the fabricated parts of the enclosure are manufactured with Stainless Steel material and Powder coated as per the standards mentioned above.

NIHVA

NIHVA Technologies Pvt. Ltd.

EPDM- Ethylene Propylene Diene Monomer Synthetic Rubber Beadings and Gaskets

The sealing of all the constituents are done by specially designed high-grade Ethylene Propylene Diene Monomer beadings and gaskets. Their excellent sealing properties ensure the enclosures 100% leak proof. Their excellent resistance to heat, oxidation, ozone and weather aging due to their stable, saturated polymer backbone structure. This ensures 100% water resistant enclosures.

Motors with remote control and push button operations

- Rack and Pinion arrangement
- Screw gear motor
- Single Phase Bi directional motor
- Speed: 10- 12 m/min
- Power supply :230V Ac,50 hz
- Power: ¼, ½ hp
- Rotating release Device: Available
- Pinion: Z16/Module 4/Module 6
- SS304 Mounting Base Built in control board.
- Operation: Through Remote control.

NIHVA
NIHVA Technologies Pvt. Ltd.

4. One-component adhesive sealant as per ASTM C920 Technical specification:

Chemical base	1-C polyurethane.
Colour (CQP1 001-1)	White.
Cure mechanism	Moisture-curing.
Density (uncured) (CQP 006-4)	1.3 kg/l approx.
Non-sag properties	Good.
Application temperature	5 - 40°C (40 - 105°F).
Tack free time2 (CQP 019-1)	60 min. approx.
Open time2 (CQP 526-1)	45 min. approx.

Shrinkage (CQP 014-1)	5% approx.
Shore A-hardness (CQP 023-1 / ISO 868)	40 approx.
Tensile strength (CQP 036-1 / ISO 37)	1.8 MPa approx.
Elongation at break (CQP 036-1 / ISO 37)	500% approx.
Tear propagation resistance (CQP 045-1/ ISO 34)	7 N/mm approx.
Glass transition temperature (CQP 509 -1/ ISO 4663)	-45°C (-50°F) approx.
Thermal resistance (CQP 513-1)	Short term 1 day 1 hour 90°C (195°F) 120°C (250°F) 140°C (285°F).
Service temperature:	-40 - 90°C (-40 - 195°F) c

Sensors

- Rain Sensor

  Rain water sensor detects the water and signals the automation to close the roof so that to avoid water entering inside the premises.

- Limit Switch

  Limit switch is a switch operated by the motion of a machine part or presence of an object. They are used for controlling machinery as part of a control system, as a safety interlock.

Operation Of Motorized Retractable Roof :

NIHVA
NIHVA Technologies Pvt. Ltd.

- Roof will be operated with remote by push button provided on it with inching mode by which user gets flexibility to open roof on different lengths up to collapsed length achieved & same can be achieved while closing of roof .

- **Polycarbonate transparent / Embose of 3mm is considered as user needs roof in special case for covering area from sunlight.**

USP of NIHVA Retractable Roof:

NIHVA
NIHVA Technologies Pvt. Ltd.

- Non-Corrosive materials like Aluminium Alloy Profiles, Stainless Steel Fasteners, Rubber sealing and Polycarbonate Sheet Fillings.
- Rain sensor or water detection sensor is a device that detects rain/water and signals the roof the retract.
- Complete Enclosure with doors, Locking systems, Rubber seals, etc. Lightweight solution for Terraces, Roofing systems, Pools, Balconies, Patios, Smoking Zones and a lot more applications.
- Completely Water proof and can be Air Conditioned if the area is enclosed.
- It can be easily Re-Located within a short time by our trained technical personal.
- No need of any license from municipal authorities for terrace Roofing's since it's completely retractable.

Scope of Work:

NIHVA
NIHVA Technologies Pvt. Ltd.

NIHVA Scope of Work

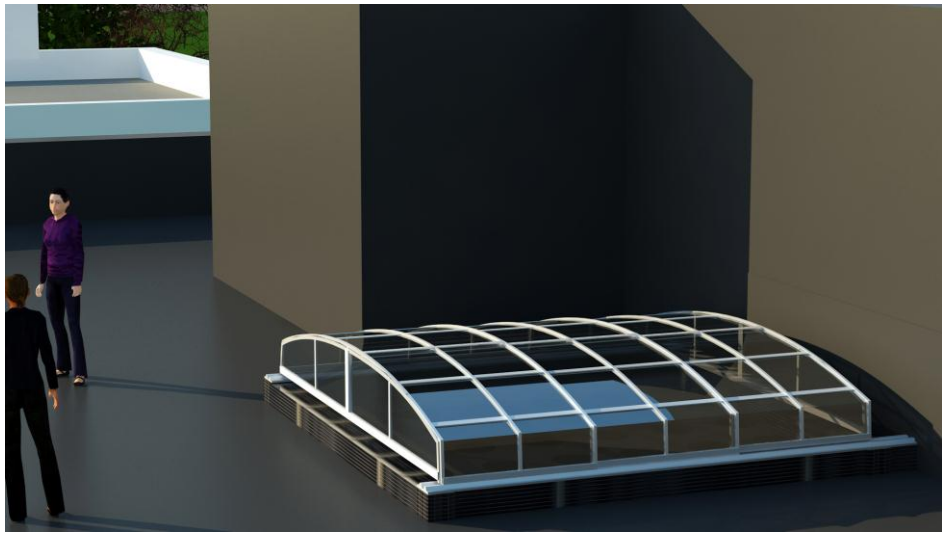
- Design, Manufacturing, Supply & installation of Automated Sliding roof
- Roof will be Fitted From Outside of existing Cutout
- Training to 2-3 Client Representative for Easy & safe working operation of roof
- Our team will visit for discussion of DAP

Client scope of Work :

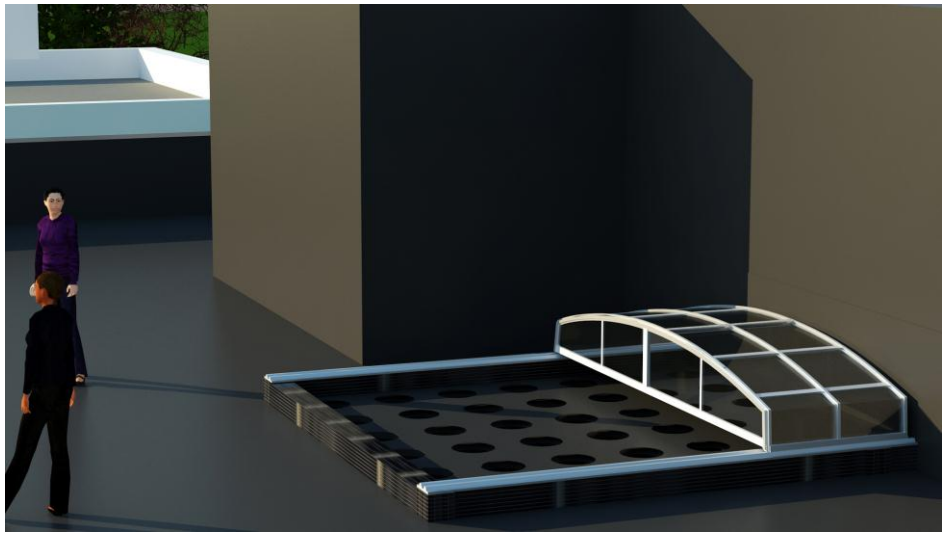
- Scaffolding for installation as roof is to be installed at some height where site is closed from above
- The dimension is considered on the basis of CAD file shared and it only shows cut-out measurement (not the actual roof dimension).
- In Case of any changes in the layout as per site requirements after the shipment of material , the additional quantities and the wastage will be to customer account.
- 3 phase electrical connection point for input supply of sliding roof.
- Civil/structural changes required on site is in client's scope.

Automated Retractable Roof Reference Images

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.



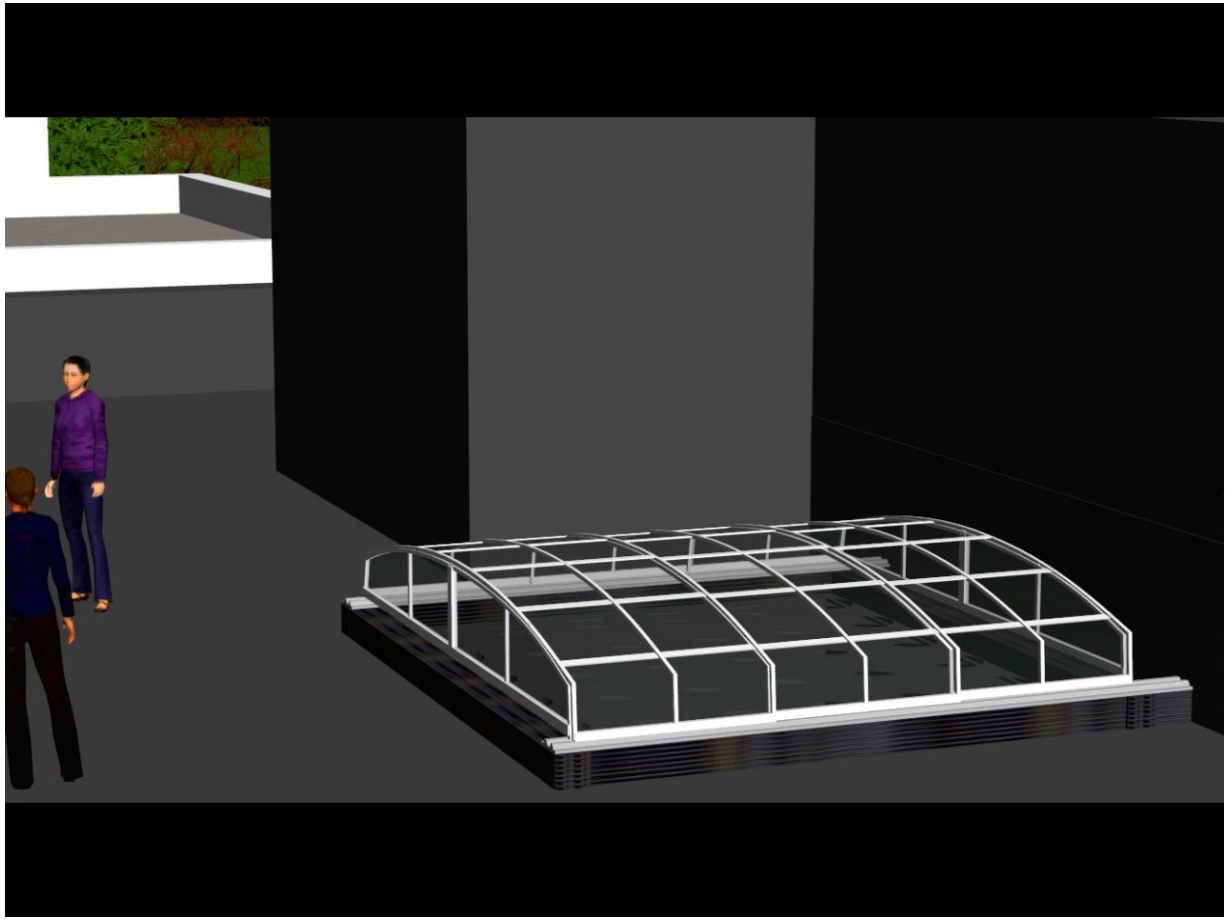






Video For Retractable Roof

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.



NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

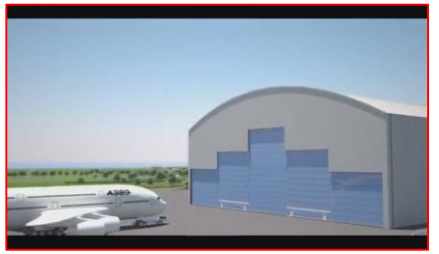
Industries Served



Automobile



Logistic & warehouse



Aerospace



Pharma Industries



Builders and construction companies



shipyards



Food Industries



Automation & Robotics

NIHVA

NIHVA Technologies Pvt. Ltd.

NIHVA Major Client List





NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

NIHVA Occurrences – Pan India

WEST

- Pune
- Mumbai
- Nashik
- Nagpur
- Indore
- Ahmedabad
- Vadodara

NORTH

- Delhi
- Chandigarh



SOUTH

- Chennai
- Hyderabad
- Bangalore

EAST

- Kolkata
- Bhubaneshwar

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Thank You

International Business Partners

ALBANY ASSA ABLOY

MEGADOOR ASSA ABLOY

Crawford ASSA ABLOY

besam ASSA ABLOY

ACO

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

NIHVA
NIHVA TECHNOLOGIES PVT. LTD.